Algorithm 1 Initialize the structure

DATA: Level L at depth d, Base number of cell N
if $d < d_{max}$ **then**
  Compute N_l and η_l
  for grid G in Level **do**
    Breakdown G in in grid of N cells G_N
    for each G_N **do**
      Compute $\overline{\nabla U_{G_N,-}}^2$ and $\overline{\nabla U_{G_N,-}}^2$
      Compare $\overline{\nabla U_{G_N,-}}^2$ and $\overline{\nabla U_{G_N,-}}^2$ to η_l
      if $\overline{\nabla U_{G_N,-}}^2 > \eta_l$ and $\overline{\nabla U_{G_N,+}}^2 > \eta_l$ **then**
        Refine G_N
      else if $\overline{\nabla U_{G_N,\pm}}^2 > \eta_l$ **then**
        Refine $G_{N,\pm}$
      end if
    end for
    Apply the initial condition to all child grid of G
  end for
  Apply **Algorithm 1** to the level at depth $d+1$
end if